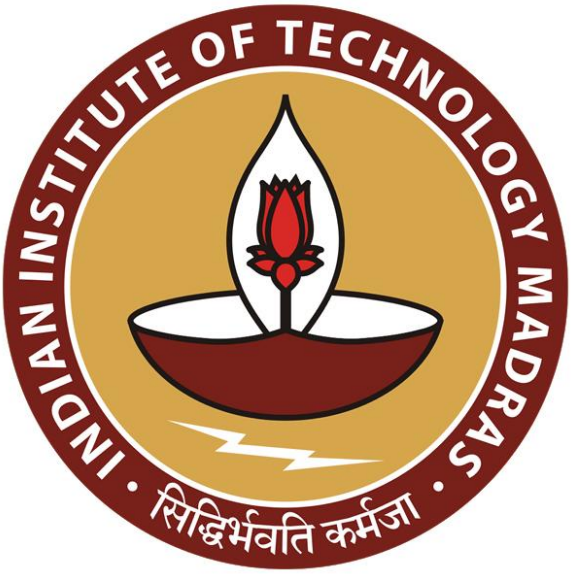
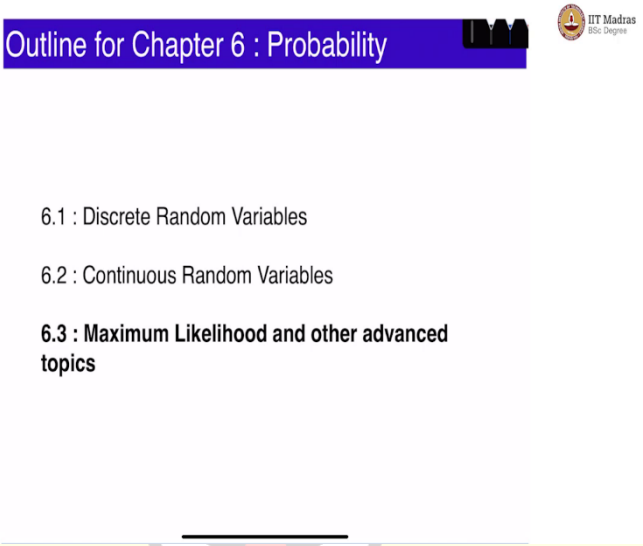


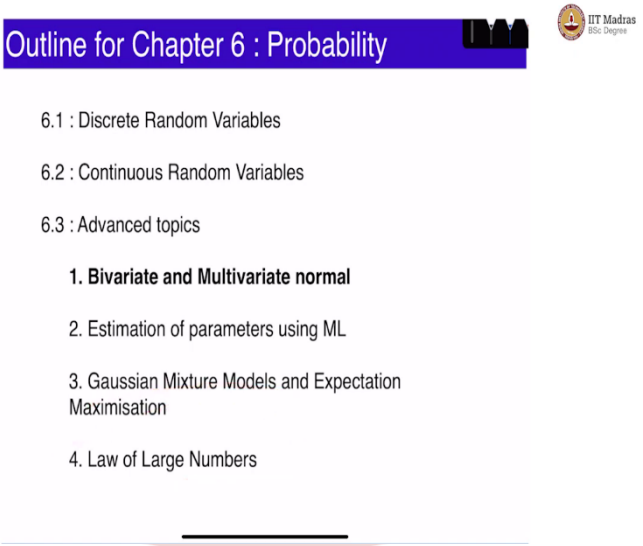
IIT Madras

ONLINE DEGREE

Machine Learning Foundations
Professor Harish Guruprasad Ramaswamy
Assistant professor, Department of Computer Sciences and Engineering
Indian Institute of Technology Madras
Chapter 6: Probability
Bivariate and multivariate normal

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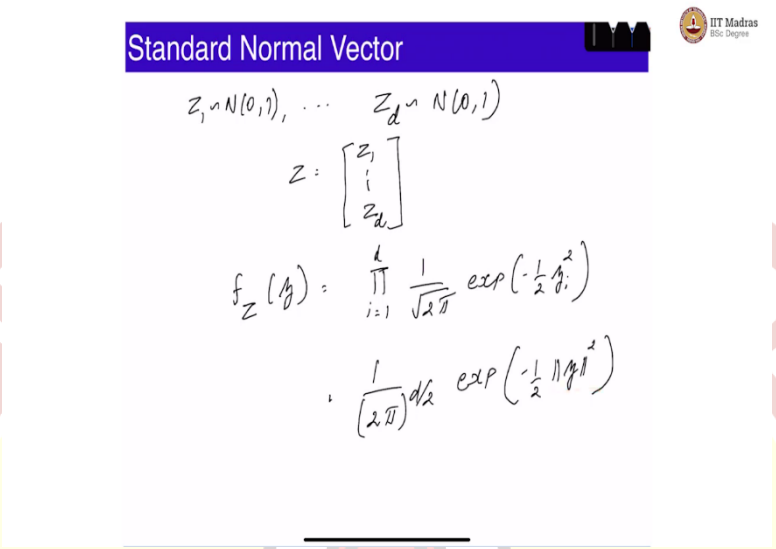




Hello, everyone, welcome to the Machine Learning Foundations course and another week of public. In this third and final week of probability we will take a look at some advanced topics, mainly focused on the maximum likelihood estimator and the multivariate Gaussian. The four main topics that we will look at this week are the multivariate normal, Maximum

Likelihood Estimation of Parameters, and the Gaussian Mixture Model and wrap up with Law Of Large Numbers.

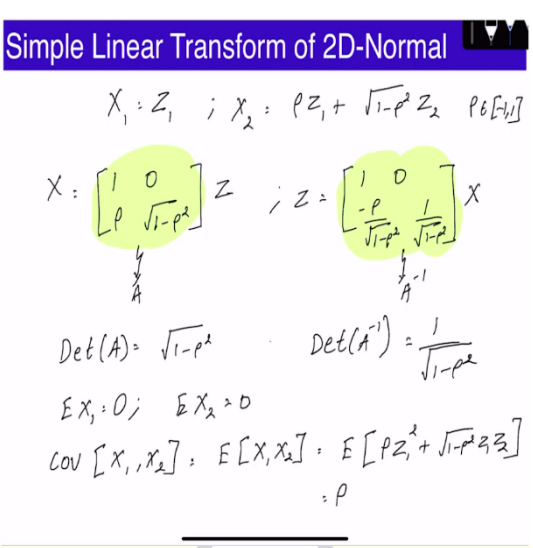
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We have seen the standard normal distribution, which is a scalar standard random variable. Let us say, we have d of those. Let us say we have Z1 distributed as normal 0,1 dot dot dot, we have d of these independent random variables. Imagine all of these put together into one vector, Z, it is a d dimensional vector, Z1 dot, dot, dot Zd. This is a d dimensional vector. What is the joint density of this random vector? What is fz offset?

Because we said that each of these components are independent, it is simply a product of the density of these components. This is product over i equal to 1 to d, 1 by square root of 2 pi, x puff - half zi square. This can be written more compactly, as 1 by 2 pi power d by 2 x puff - half norm of Z square. This Z is a vector and essentially, the product of x essentially can be written as x puff the sum of the squared of zi, which is exactly equal to nom z square. So this is an example of a normal vector.

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Now, the goal of this particular section of the course is to introduce you the multivariate normal, we will do that in gradations. We will slowly increase the complexity, little by little until you reach the full multivariate normal. The first step that you can consider is a two dimensional normal. Let us say you have Z1 and Z2 and we have this particular linear transform of this standard normal random vector Z, we will define X1 to be equal to Z1, and X2 to be equal to rho times Z1 + square root of 1 - rho squared times Z2, where rho is sum scalar, between - 1 and 1.

Because Z1 and Z2 are already standard normal vectors, you can use Z as essentially a two-dimensional standard normal vector. And we have transformed Z1, Z2 into two random variables X1 and X2, which is a linear transform X1 and X2 form a linear transform of Z1 and Z2. Why this particular linear transform? We will get to that. For now, let us just consider this linear transform.

We can write this linear transform in terms of a vector matrix operation, which is let us say X is a two-dimensional vector consisting of X1 and X2 that simply equal to 1, 0, rho square root of 1 - rho squared Z. Z is a two-dimensional vector X is a two-dimensional vector and is a 2 * 2 matrix. This is useful for us because we want to get the density of X. So, we are writing the density of X in terms of, so we are writing X in terms of Z, so that we can use the change of variable rule.

To get the change of variable rule, we need to also be able to invert this. And this happens to be an invertible linear transform. So, you can actually get that Z is simply equal to this matrix inverse times X. And what is the inverse of this matrix you can do, you can use your standard school linear algebra to get the inverse of this 2 * 2 matrix, which is simply 1, 0 - rho by, sorry about that, - rho by square root of 1 - rho squared and 1 by square root of 1 - rho square.

So, this is a 2 * 2 matrix, you can easily check that this matrix is the inverse of this matrix. So, we have X, X is equal to this matrix times Z, Z is simply equal to the matrix inverse times X. So, let us call this matrix as A and this matrix as A inverse. It will be useful for us to compute the determinant of these things recall, to get the change variable rule, we need the determinant of the Jacobian and the Jacobian happens to be exactly this, so we can take the determinant of the Jacobian.

Let us say what is the determinant of this matrix A is a 2 * 2 matrix it is really simple determinant of A is simply equal to square root of 1 - rho square. And if you know the properties are the determinant, determinant, sorry about that, determinant of A inverse is simply 1 by determinant of A determinant of A inverse this 1 by square root of 1 - rho square. Well we are just building up slowly to get the density of X.

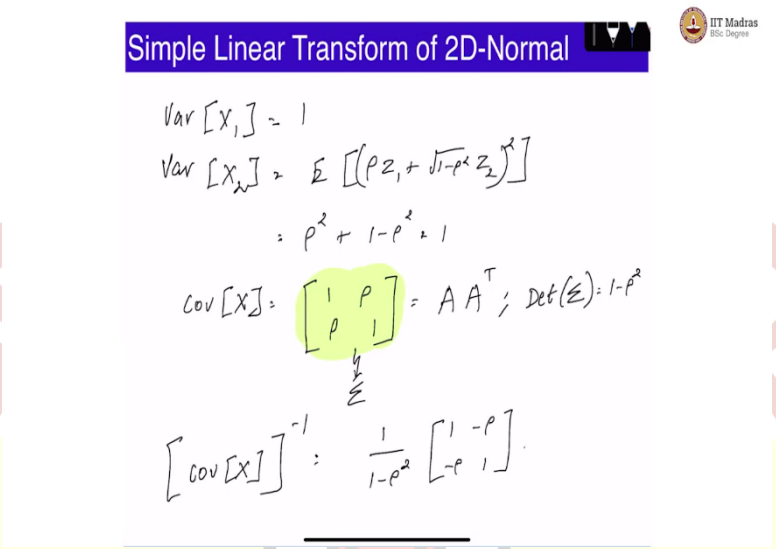
Before we go there, let us get some properties of X. So, X is a random vector, what is the expected value of X? X has two components X1 and X2, what is EX1, EX1 is simply EZ1 which is 0. What is EX2? EX2 is expected value of rho, Z1 + square root of 1 - rho squared Z2. Once again by linearity of expectation, we have EX2 is also equal to 0. So now we will get to, what is covariance of X1 , X2.

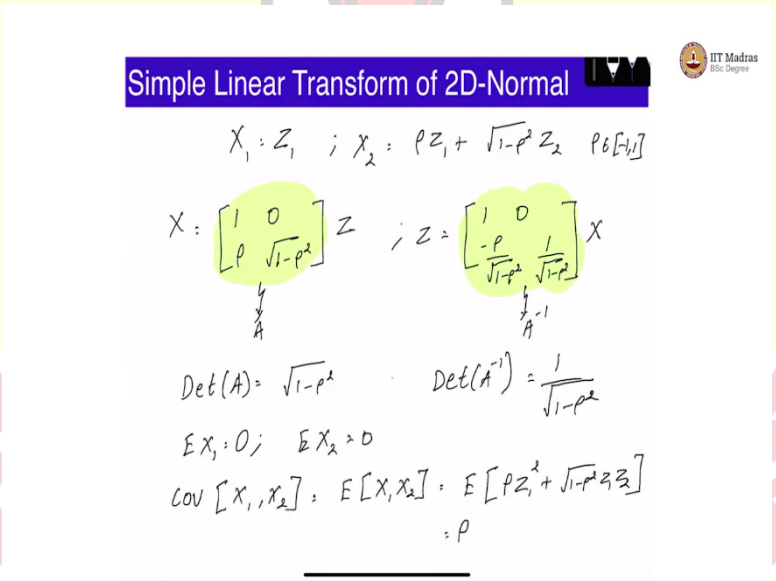
What is that? Well, that is expected value of X1, X2 - expected value of X1 into expected value of X2 because the EX1 and EX2 are 0 this is simply equal to expected value of X1 into X2. And X1 is already Z1, X2 is rho Z1 + square root of 1 - rho squared Z2, which means, that this particular term is simply equal to expected value of rows Z1 squared + square root of 1 - rho squared, Z1, Z2. Once again, I can use linear expectation to write expected value of rho Z1 squared + square root of 1 - rho squared Z1, Z2 as this expectation + this expectation.

But we know that Z1 one and Z2 are independent that is, those are our fundamental assumption. So, Z1 and Z2 are independence so expected value of Z1, Z2 is going to be equal to 0. So now we end up with expect rho with expected value of Z1 squared. And we said that

Z1 was a standard normal, so expect value of Z1 squared is simply equal to 1, so we have the covariance of X 1, X2 is equal to rho.

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And let us just complete it, and get also the variance of X1 and X2. What is variance of X1? X1 is equal to Z1, so variance of X1 is also equal to variance of Z1, which is equal to 1. What about variance of X2? Simply equal to the expected value of rho Z1 + square root of 1 - rho squared Z2 the whole square. And by simple linear expectation, this is again rho squared + 1 - rho squared is equal to 1.

So now we have variance of X1 and variance of X2 are equal to 1 and covariance of X1 , X2 is equal to rho. Put these together, we get the covariance matrix of the vector X. What is that?

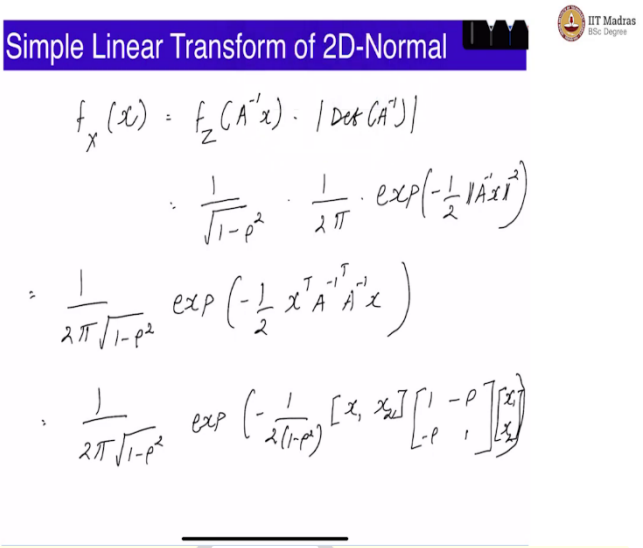
Covariance matrix of the vector X is simply given by a 2 * 2 matrix where the diagonal entries are simply the variance of X1 and X2, which is 1 and 1. And the off-diagonal entries, the symmetric matrix is going to be equal to the covariance of X1, X2 is going to be equal to rho.

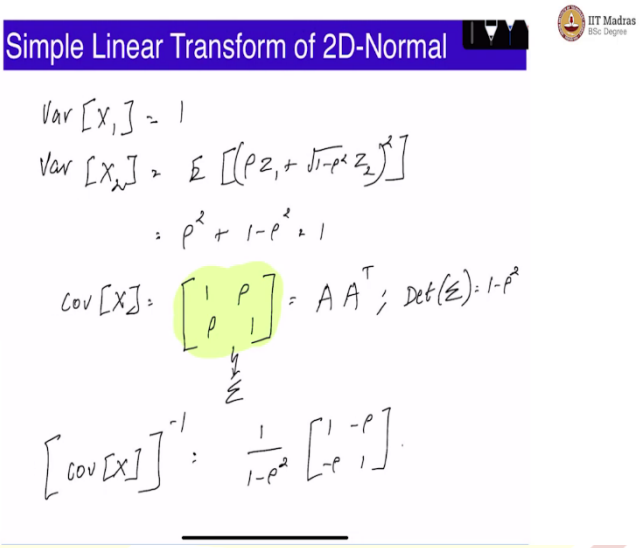
And here is a simple check for you, this is exactly equal to A times A transpose. A recall a was the linear transform matrix here, you can simply do a simple check here and check that A and A transpose is exactly equal to 1 rho, rho 1. For, now, just do a simple check and check that out. And this helps us because what is determinant of sigma? Well, we will see why we will call this matrix as the sigma.

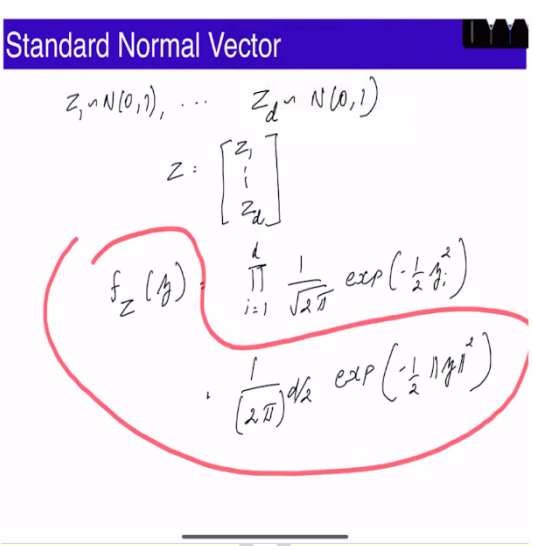
We will call name this this matrix. We will call this as sigma. What is determinant of sigma? It is simply equal to 1 - rho squared. We will also need the inverse of the covariance matrix. We will get that also. I am right now just putting the ingredients for the recipe. We will cook all of these together to get the density of X soon. But what is the covariance? X inverse. This is also 2 * 2 matrix you can pretty easily invert this it is going to be equal to 1 by 1 - rho squared, 1 - rho - rho 1.

What is the determinant of this? It is simply going to be 1 by determinant of sigma, which is 1 by 1 - rho squared. This follows from the properties of determinant, determinant of A is equal to 1 by determinant of A inverse. So now we have all the ingredients to answer the question, which is, now we started off with the random standard normal random vector Z, and we did a linear transform to get random vector X.

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Now what is the density of this new random vector X? What is f x of x? This is because there is invertible linear transform gave us the change of variable rule to get fx of x is simply equal to FZ of A inverse x into the determinant of the Jacobian, which happens to be exactly equal to A inverse, so determinant of A inverse.

Now let us be noted A inverse is equal to one by square root of 1 - rho squared and fz of A inverse x is 1, we will write that first, 1 by square root of 1 - rho squared and Z is a, this is a two-dimensional standard normal vector so that is going to be 1 by 2 pi. Because two-dimensional root 2 pi squared is 2 pi, 1 by 2 pi exp puff - half norm of A inverse x the whole squared. So, this is just, I have substituted the A inverse x in the density expression for Z standard normal vector Z.

We call this is the density of the standard normal vector of dimension d, because d equal to 2, this becomes 1 by 2 pi. And just substituted A inverse x for Z. Now, we will expand the normal vector norm of V squared is simply equal to B transpose V, you can use that property to expand this further, let us expand this further. I need a little bit more space, so I am generating that by going to the left here. So, you have 1 by 2 pi square root of 1 - rho squared x puff - half x transpose A inverse transpose A inverse x.

So, because norm of A inverse x, the whole squared is simply A inverse x transpose A inverse x, so A invese x transpose is x transpose A inverse transpose which is simply properties of linear algebra, and you get this particular expression. Now let us look at this particular matrix A inverse transpose A equals. What is that? Well A inverse transpose A

inverse is simply equal to the inverse of the covariance matrix sigma, that is exactly equal to sigma inverse.

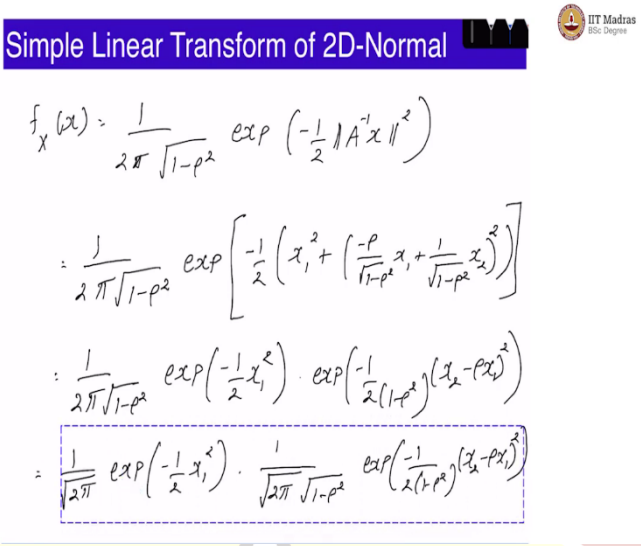
We will put that in here, so that is 1 by 2 pi square root of 1 - rho squared, x puff - half, we will expand x transpose as x1, x2 and A inverse transpose A inverse is simply equal to sigma inverse, which is 1 - rho - rho 1, and you do have a scalar, which is equal to 1 by 1 - rho squared, we will put that in for the sigma inverse here, so that it is easy to 1 by 2 into 1 - rho square.

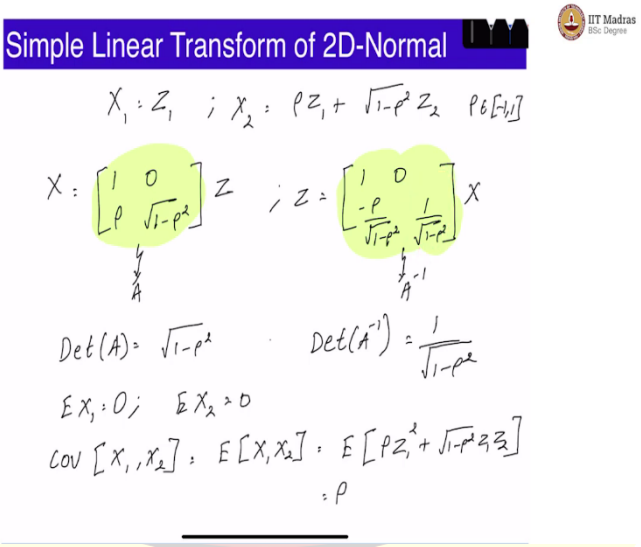
So why is that? Because A inverse transpose A inverse is simply exactly equal to sigma inverse. Sigma is equal to A transpose, so sigma inverse is going to be A inverse transpose A inverse, which is exactly what we have here. And sigma inverse we have already calculated is simply equal to 1 by 1 - rho squared, 1 - rho - rho 1. So that is why you have 1 by 1 - rho squared, 1 - rho - rho 1, we will just complete this x1, x2 and then you need to just close the parenthesis.

So, this is the expression for the density of the two-dimensional random vector x. This is an example of the multivariate normal or to be more precise, this is an example of the bivariate normal. Still not clear, what special is going on here, but this is just a very special case. And the reason why this is a special case is that this is particularly easy to analyze.

We will do some more analysis of this before going ahead and generalizing to the general multivariate normal. Before we simplify this, let us go take a step back into this expression and simplify it in a different way. So, we will start off with this expression, but go ahead and simplify it in a different way.

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Which is fx of x is equal to 1 by 2 pi square root of 1 - rho squared into x puff - half norm of A inverse x the whole square. This is the expression, we will just simply it in a different way. How do we do that? Well, you go back to the expression of A inverse. What is A inverse A inverse was exactly this matrix. Remember?

A inverse was this matrix, so A inverse, we can just compute A inverse x as a vector, and the norm squared of that vector is simply the sum of the squares of each of these elements. So, A inverse x, the elements of vector A invers x are simply x1, and - rho x1 by square root of 1 - rho squared + x2 by square root of 1 - rho squared. So, let us just compute the norm squared of that particular vector that is going to be, the first term is going to be the same 1 by 2 pi square root of 1 - rho squared into x puff - half.

Norm of A inverse x, the whole squared is going to be x1 squared + - rho by square root of 1 - rho squared x1 + 1 by square root of 1 - rho squared x2, the whole square. I am just going to use a square bracket for the x so that it is a little bit easy to see. Recall A inverse x is simply a vector given by x1 and - rho x1 by square root of 1 - rho squared + 1 by square root of 1 - rho squared x2. So, the norm of that is norm squared of that is simply given by this.

So now let us expand this a little more and let us simplify this a little bit more. This is 1 by 2 pi square root of 1 - rho square. And instead of the x being in the sum here, we will break that down into product of two terms. The first term is simply going to be x puff - half x1 squared, the second term is going to be just essentially, this term, which is x puff - half into x puff - half to 1 - rho squared into x2 - rho x1, the whole square.

I have just more the sum here. So, you get 1 - rho squared, because you can take 1 - square root of 1 - rho squared outside, we have square that will become - 1 by 2 into 1 - rho square. So, I have just moved terms, I have done nothing big here. Why have I done that? Well, I am going to move on even further and I am going to split 1 by 2 pi square root to 1 - rho squared as 1 by square root of 2 pi, and 1 by square root of 2 pi into square root of 1 - rho square.

Why am I doing all this? We will see pretty quickly, but I am going to do that first. This is 1 by square root of 2 pi x puff - half x1 squared into 1 by square root of 2 pi into square root of 1 - rho squared x puff - 1 by 2 into 1 - rho squared x2 - rho x1the whole square. I have done nothing I have just moved terms here and there to get into a convenient form. Why have I done all this? Let me just copy paste, copy this expression and paste it in the next page to do some more to wrap up the analysis.

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Simple Linear Transform of 2D-Normal

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}x_1^2\right) \cdot \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} \exp\left(\frac{-1}{2(1-\rho^2)}(x_2-\rho x_1)^2\right)$$

$$= f_{X_1}(x_1) \cdot f_{X_2|X_1}(x_2|x_1)$$

$$N(x_1 \mid 0, 1) \qquad N(x_2 \mid \rho x_1, 1-\rho^2)$$

$X_1 \sim N(0,1)$

$X_2 | X_1 = x_1 \sim N(\rho x_1, 1-\rho^2)$

Simple Linear Transform of 2D-Normal

$$f_X(x) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2}\|A^{-1}x\|^2\right)$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left[\frac{-1}{2}\left(x_1^2 + \left(\frac{-\rho}{\sqrt{1-\rho^2}}x_1 + \frac{1}{\sqrt{1-\rho^2}}x_2\right)^2\right)\right]$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2}x_1^2\right) \cdot \exp\left(\frac{-1}{2(1-\rho^2)}(x_2-\rho x_1)^2\right)$$

$$= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}x_1^2\right) \cdot \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} \exp\left(\frac{-1}{2(1-\rho^2)}(x_2-\rho x_1)^2\right)$$

Simple Linear Transform of 2D-Normal

$$f_X(x) = f_Z(A^{-1}x) \cdot |\det(A^{-1})|$$

$$= \frac{1}{\sqrt{1-\rho^2}} \cdot \frac{1}{2\pi} \cdot \exp\left(-\frac{1}{2}\|A^{-1}x\|^2\right)$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2}x^T A^{-1^T} A^{-1} x\right)$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\begin{bmatrix} x_1 & x_2 \end{bmatrix}\begin{bmatrix} 1 & -\rho \\ -\rho & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right)$$

So, I have no, this is fx of x I am not going to write fx of x on the left side that is understood. This is fx of x. Now, we have two terms. The first term is this and the second term, let us use a different color, is this. Let us try to understand each of these two terms separate. So, this is fx of x. X is a vector whose competence are x1 and x2 it is a two-dimensional vector, fx of x is this.

But we already know that the joint distribution of let us say two random variables can be written let us say fxy, fx , y can be written as fx of x into fx given sorry fy given x of y given x. So, on that basis, we can write down the joint distribution of these two random variables x1 and x2 as f x1 of x1 sorry f x1 of x1 into f x2 given x1 of x2 given x1.

This is always possible. Pick any random vector x1, x2 I can always write the joint distribution into product of these two. Now you should guess why the, I have split this into two terms. You can easily see this corresponds to this, you can see this is the valid density function. Why is that? This exactly corresponds to the density function of a standard normal. You can see 1 by square root of 2 pi x puff - half x1 squared, this exactly corresponds to a standard normal with mean 0 and variance 1.

That means, that this term here must correspond to x2 given x1. So here, note that x1 is a constant. So, in this particular expression you should view x1 as a constant and you should just view this as a density or x2. If you do that, you can see you have x2 - rho x1 the whole square that kind of given that rho x1 as a constant, this essentially is saying that this corresponds to a normal of over x2 with mean given by row x1.

And what is the variance? The variance is 1 - rho squared and that does make sense because your square root of 1 - rho squared. So, put these together, you get that the first term, which means that the first term here, this corresponds to the normal with me of x1 eval with the let us say mean 0 and variance 1. What about this term? This term here corresponds to the normal of x2, but what is the mean, the mean is given by rho x1, and the variance is given by 1 - rho squared.

So put these together, we get the x1, the density of x1 is normal with mean 0 variance 1 that does make sense because x1 was exactly equal to z1 that is how we defined it. And x2 given x1 is a normal with mean given by rho x1 and variance given by 1 - rho square. So, let us write this expression down, which is x1 I think this is let us do that way, x1 is distributed as

normal with mean 0 , 1, and x2 given x1 is equal to x1 distributed as normal with mean rho x1 and variance 1 - rho square.

So, this is the power of breaking the expression, the joint density of x into these two things. We can any joint density, you can always break into these two. The advantage is normal, it seems that both of these also are not. It might not always be possible for some other joint density, if you can break it into like this into two distributions like this, both of these may not be normal, and that is one of the properties of the multivariate normal. If you break it down like this, each of these components are also going to be normal.

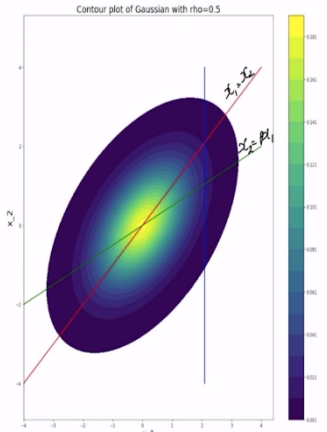
We will not prove that just understand this for this particular special case. Once again, immediately, you should ask the question, well, this is one way to break down x1 and x2 that is x, I can break down the joint density of x1 as the marginal of x1 into the conditional of x2 given x1. But I can also do the other way around. I can make it the marginal of x2 into conditional of x1 given x2, that is also a perfectly valid way to do it. And can we do that?

The answer is, yes, you can do it is a simple exercise to think about what will happen if you do that? I mean, if you write x1, instead of x1 you put x2 and you replace x1 and x2, what will happen, that is a good question to ask. The answer is kind of staring at you there through this expression. This is also a valid expression for the joint density, we just derived it. And you can see that in this expression there is a symmetric over x1 and x2.

You could have swapped x1 and x2, and you would have gotten the same answer. So, this indicates that x1 and x2 you can swap and you will get the same answer. At least if you have a statement for x1, x2 you can swap it will hold for that also. So, it means that in this particular expression we can swap it, swap 1 and 2, in subscripts.

If you do that, it means that x2 is also distributed as normal with mean 0,1 and x1 given x2 equal to x2 is distributed as normal with mean rho x2 and variance 1 - rho squared. This is an exercise for you. Do not, you do not need to take it for granted from me, you can just take this expression for fx of x and try to break it down into fx2 two into fx1 given x2. You can do that and check whether that holds. That is a simple exercise for you to do when you are free.

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Simple Linear Transform of 2D-Normal

$$f_X(x) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2}\|A^{-1}x\|^2\right)$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left[\frac{-1}{2}\left(x_1^2 + \left(\frac{-\rho}{\sqrt{1-\rho^2}}x_1 + \frac{1}{\sqrt{1-\rho^2}}x_2\right)^2\right)\right]$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2}x_1^2\right) \cdot \exp\left(\frac{-1}{2(1-\rho^2)}(x_2-\rho x_1)^2\right)$$

$$= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}x_1^2\right) \cdot \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} \exp\left(\frac{-1}{2(1-\rho^2)}(x_2-\rho x_1)^2\right)$$

Simple Linear Transform of 2D-Normal

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}x_1^2\right) \cdot \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} \exp\left(\frac{-1}{2(1-\rho^2)}(x_2-\rho x_1)^2\right)$$

$$= f_{X_1}(x_1) \cdot f_{X_2|X_1}(x_2|x_1)$$

$$N(x_1|0,1) \qquad N(x_2|\rho x_1, 1-\rho^2)$$

$$X_1 \sim N(0,1)$$

$$X_2|X_1 = x_1 \sim N(\rho x_1, 1-\rho^2)$$





So now let us go ahead and visualize this particular density. We have the fx of the expression for the density of x. So, that means it is a function of two variables x1 and x2. It is a function of two variables. How do we visualize functions of two variables, the standard way to visualize functions of two variables is through either contour plots or heat maps. So that is what I have done here.

I visualize the joint density over x1 and x2 with the contour with the heat map over x1 and x2. In this case, I have just said rho to be equal to 0.5. So that is the I have said rho to be equal to 0.5 and visualize this. You can see that it is an ellipse with a slant, so there is a slant at this angle here.

So, I have several lines here. Let us just look at the red line. Red line simply corresponds to x1 is equal to x2. And the green line corresponds to x2 is equal to rho x1. So, we can see that the red line is at 45 degrees, the green line is not at 40 degrees it is less than 45 degrees, because x2 is equal to rho x1 corresponds to align with angle I do not know it is less than 45 degrees.

So, it is x1 equal to x2 and x2 equal to rho. Why do we have these two lines? Because x2 two given x one is normal with mean row x one and one - one squared variance. So, what does this mean? Let us go one step further. And instead of having a contour plot, let us just sample points from this density, so we can this is x1, x2 is a two-dimensional random variable, I can sample points from this density, and get the points.

So, if you did that, this how sample, I have sampled 10,000 points from this two-dimensional distribution, I have gotten these points. I mean, each dot here corresponds to a sample point x1 , x2. Now let us look at the expression for this. So, we say that x2 given x1 is equal to x1 is also normal, with mean row x1 and variance 1 - rho squared. So, that means, that if I say x1, fix x1 to some value, let us say x1 I fix it to this value then x2 happens to be normal with mean given by rho x1.

How do we do that? In practice, I mean, we cannot set exactly equal 2 or any value because A is a continuous random variable setting anything exactly to a value is always going to be 0, I mean, number of samples exactly equal to 2 is going to be exactly 0. But what you can do is you can set a small neighborhood around 2. And what I have done here in the next slide is just take samples where x1 is close to 2.5. So, this, I have just set x1 to be equal to 2.5, not exactly 2.5, I think 2.49 to 2.51.

I have taken all samples of the full set, where x1 is between 2.49, 2.51 and I have just plotted that here such so you can see there is some kind of width here, but the width is small and then you can ignore that. So, you can see that this also happens to have some kind of it is mostly concentrated or at the center here. And as the, as it goes away from the center, you have lesser and lesser points.

And where is that center? That center exactly coincides with x2 is equal to rho x1, because x2 given x1 is normal, with mean given by rho x1 and variance given by 1 - rho squared. Similarly, you can example you can do the same thing for let us say x1, you can set x1 to be

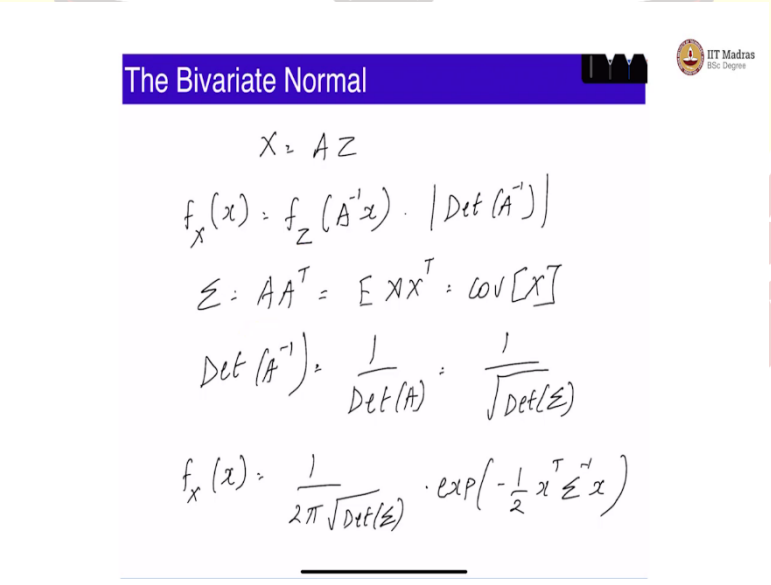
the neighborhood of - 2. And here also you can see that x2 is mostly in this around - 1 and as x2 goes farther away from - 1, there are lesser and lesser samples.

So, x2 is most likely to be equal to around - 1, as the farther it goes from - 1, the less likely it becomes. So that is why we have lesser and lesser samples farther from - 1. So why is that - 1 because when x2, x1 is equal to - 2, rho x1 is exactly equal to - 1. So, this is what the conditional distributions capture. x2 given x1 is also normal with mean given by rho x1.

In fact, you can do the other way around also x1 given x2 is also normal with mean given by rho x2. So that is another interest. So, we had excusable rho x1 as this green line. What will x1 equal rho x2 line look like? It will not be the same line, x2 equal to rho x1 this line means that x1 equal to rho x2 is going to be a different line.

In fact, it is going to be aligned like this. And you can set you can freeze your x2 value and look at how the x1 varies, you will get a very similar looking curve, just that instead of vertical lines, you will get horizontal lines, horizontal scatter plots, that is the only difference. So that is how you can interpret joint two-dimensional normal distribution. So, this should give you some idea of this particular example.

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Now we will go back to the situation. Why this particular example why mean, we the way we got this particular random variable X was we set X to be equal to some matrix times Z. We set X to be equal to A times Z where A was the strange matrix given by 1,0 rho square root of

1 - rho squared beta. So, we had this very strange matrix and we said this was the matrix. What is special about this matrix? Really, there is nothing special about this matrix.

You can just not have this matrix you can use any matrix A, but it should be invertible, though, because we are going to have to use the change a variable rule. So that means that x equal to A, if A is any invertible matrix, we can actually apply the idea here and it will still be valid, by the variant.

Let us go ahead and do that. So, if X was AZ where A was any invertible 2 * 2 matrix, it can be anything. In that case, what is fx of x going to be? Well, lot of things go through the same way, nothing much really changes, but we will do that anyway. Fx of x is simply going to be equal to fz of A inverse x into the absolute value of the determinant of A inverse.

Once again, let us have sigma to be equal to A, A transpose and you can also check that this is exactly equal the expected value of XX transpose, which is the covariance of the vector X. You can have determinant of A inverse is equal to 1 by determinant of A which is equal to 1 by square root of determinant of sigma.

Why is that? Because A and A transpose have the same determinant and the determinant has the multiplicative property. Determinant of sigma is equal to determinant of A, determinant of A transpose, which means that determinant of sigma is equal to determinant of A squared, which means the determinant of A is simply the square root of determinant of sigma.

Why am I doing all this? Because I can just substitute determinant of A inverse here in this expression. That is the reason why I am doing all this. Let us just substitute the expression for determinant of A inverse in this expression, we get fx of x simply equal to 1 by 2 pi square root of determinant of sigma into x puff - half x transpose sigma inverse.

Why do you get x transpose sigma inverse x? Well, you, you will get x transpose A inverse transpose A inverse x, but that is exactly equal to sigma inverse. All this looks pretty much similar to the case where we had that strange matrix. Nothing really, nothing changes really, it is exactly the same thing, just that we have the more general expression here for determinant of sigma and sigma inverse here.

One thing to notice, A, that actually does not apply come in the density of x. The only way the matrix A occurs, the density of x is through sigma. You could potentially imagine two

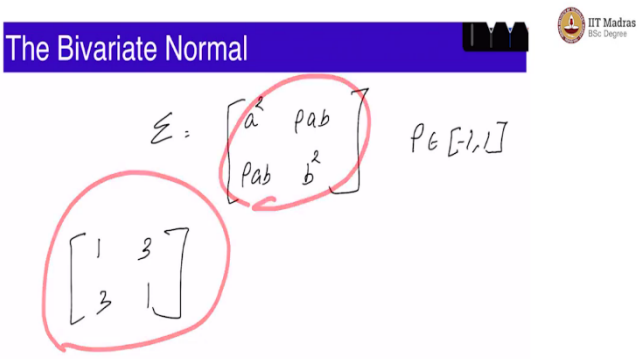
different linear transformations A having the same sigma, is possible to have two distinct linear transformations A for which AA transpose is the same. And both of those linear transformations will have the same density.

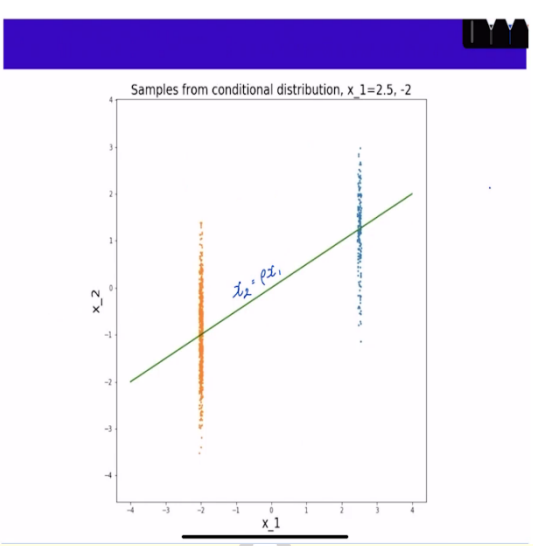
Mind you, they are not the same variable, but they are the same density. If you want an example, let us take just a one-dimensional example. Let us say Z is a standard normal random variable, you can imagine two random variables, Z and - Z. Clearly, these two are different random variables, but Z and - Z have the same density. Why is that? Because that happens to be symmetric distribution, so Z and - Z have the same density.

So, on the same logic, you can have two, the only way the density of x depends on A is through sigma. So, A can be any 2 * 2 metrics. Of course, it should be invertible 2 * 2 metrics. But does that mean the sigma can be any, sigma is also a 2 * 2 sigma matrix, sigma is AA transpose, can sigma be any 2 * 2 matrix? The answer is no, because sigma has to be written as AA transpose.

For example, in the case of scalars, sigma would be something like A squared. A squared any, any number cannot be written as A squared only positive numbers can be written as A squared. In a similar way, any matrix cannot be written as AA transpose only what are called positive definite matrices can be written as AA transpose. If especially A is invertible only positive definite matrices can be written as AA transpose. I hope you all you all know what positive definite matrices are.

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But even if you do not know in the case of 2 * 2 matrices, positive definite matrices happen to have this kind of structure. So, the diagonal entries can be any positive number. Let us say a squared and b square. the off-diagonal entries on the other hand cannot be anything. They have to be between ab and - ab. So, let us write that down as rho times ab rho times ab.

So, a and b can be anything. So as long as the, so diagonal entries are always going to be positive, and the off-diagonal entries have to between - ab and + ab, so rho must belong to - 1 to 1. So, you could, in principle, pick any a , b, and you can write down a covariance matrix like this, so that would be a valid 2 * 2 covariance matrix.

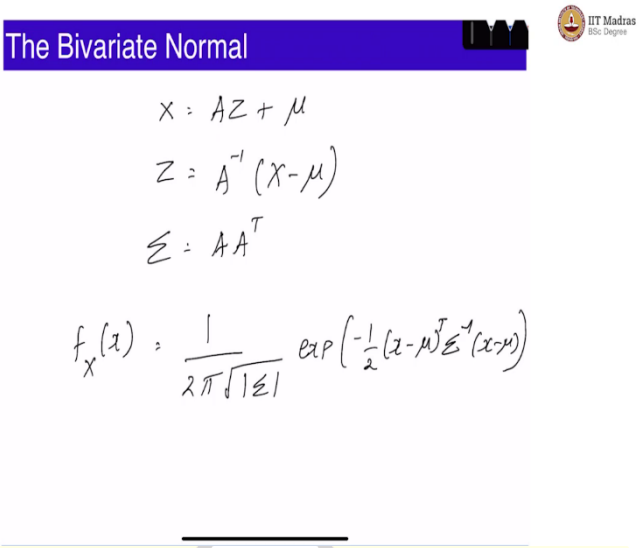
Clearly, you can see that all 2 * 2 matrices cannot return as covariance matrix. Here is an example of a covariance matrix, which cannot be, for example, let us say 1,3,3,1. You cannot express this matrix in this form. Why is that? Because rho seems to be greater than 1, so that is allowed. So not all 2 * 2 matrices are positive matrixes. So, and all positive definite 2 * 2 matrices can be written in this form.

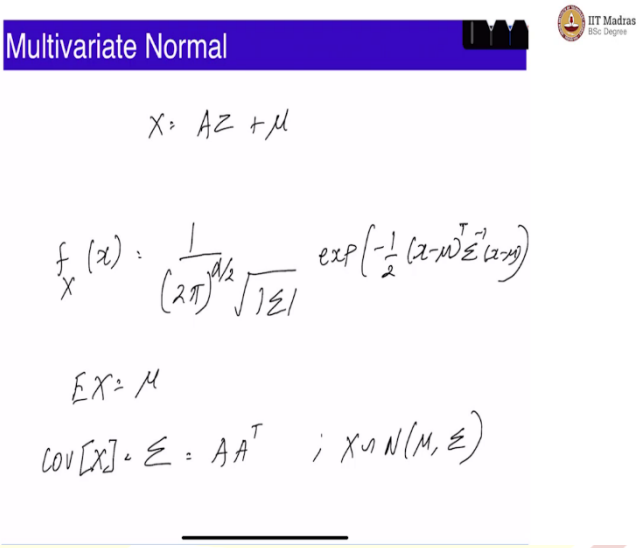
It is really useful for us later when we do some analysis, but that is all you need to know right now. Right now, we started off with one particular covariance matrix sigma, which was 1 rho, rho 1, so you can see that that is a special case of this thing, where you have both a and b to be equal to 1. If you set a equal to b equal to 1, you get 1 rho, rho 1, which is what we had in the first example. So that is a very special case.

But in general, it can be anything. It can be a squared rho ab, rho ab, b squared. So, that is a general format of a 2 * 2 covariance matrix.

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So now, we will just generalize it to the full extent of bivariate normal, because right now, we have been only doing linear transforms. That means, that the mean is always going to be 0. What if you wanted a bivariate normal whose mean is not 0? In that case, you wanted to also have that effect, so we can write x to be equal to A times Z + a constant vector mu, mu is a two-dimensional vector you can say x is equal to AZ + mu, this is also invertible transform, because you can write Z is equal to A inverse into X - mu.

You can have sigma is equal to AA transpose in exactly the same way, and you can get the density of this random vector X also in the same way. The only thing that changes is that instead of x you have x - mu, that is the only thing that changes the 1 by 2 pi square root of

determinant of sigma x puff - half x - mu transpose sigma inverse x - mu. So, this is the general bivariate normal expression.

Now, we come to the point that there is no nothing special about two dimensions, you can just as well extend this to higher dimensions as well. You can have a general d dimensional set and you can have a general d * d transform. So, you have here a 2 * 2 transform a two dimensional random vector Z standard normal random letters Z, we can have a d dimensional standard normal independent random vector Z and we can have a d * d matrix A.

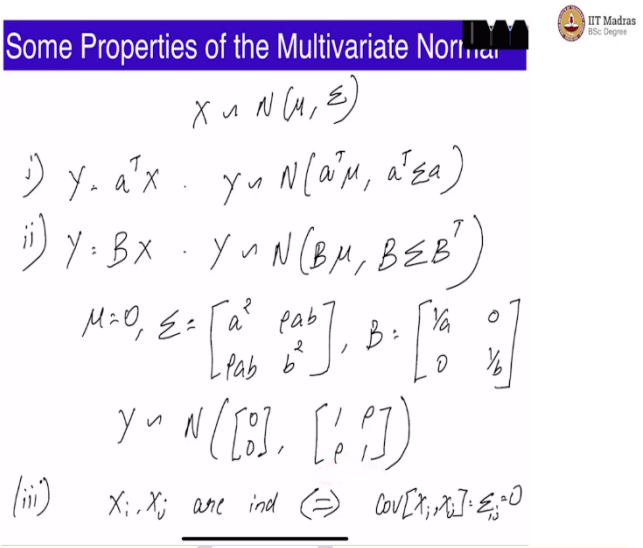
And pretty much everything will go through the only change that will happen is that you can have execute AZ + mu exactly everything is the same just that the sizes of all of these are different now. A is a d * d matrix Z is a d * 1 matrix mu is also a d * 1vector. What happens is the density? Density looks almost the same. The only change is that instead of 2 pi, you have 2 pi power d by 2,that is the only change that happens everything else is exactly the same.

So, square root determinant of sigma x puff - half x - mu transpose sigma inverse x - mu. What is the covariance of this particular random vector? Expected value of x is equal to mu and covariance matrix of x is equal to sigma. You can compute the covariance of this matrix using just the expression of x because you know that the covariance matrix of the standard normal vectors Z is simply the identity matrix.

So, you can use that to get covariance matrix up this vector x matrix sigma, which is equal to AA transpose. So, to get the density of the random method X you need only two parameters, which is the mean vector mu and the covariance matrix sigma. So, this is compactly written as X is distributed as normal with mean mu and covariance matrix sigma. So, this is a standard notation used for multivariate normal.

This is perhaps the if you are planning any carrier in machine learning and data science, this is an expression which you are going to see so often that you are not even going to be ever surprised by this expression, whenever you see things happening. So, this is this multivariate normal with mean, mu, mean vector mu on covariance matrix sigma. These, this is of size d * 1, this is of size d * d.

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So, there are several properties the multivariate normal has. We will not go into the full detail here, but some important properties I will just list here. Let us say if X is distributed as normal with mean mu, and covariance matrix sigma then first property is that let us say you have another random variable Y, it is a scalar random variable is equal to a transpose x, let us say a is some fixed vector, constant vector, and Y is a scalar random variable, which is simply dot product of a transpose X, what you can show is that Y is also a normal and a variable scalar just scalar random variable.

So, the mean is given by a transpose mu and the covariance is given by a transpose sigma a. You can in fact, extend this further to have vector transformations, you can have an let us say a vector transform b, let us say Y is equal to B times X, B is some d * d matrix. What is Y going to be Y is also going to be now a multivariate normal, because Y is a d * b Y is a d-dimensional vector X is also a d-dimensional vector Y is a normal with the mean given by the vector B times new and covariance given by B sigma B transpose.

So, these are reasonably easy to derive, I am not going to do that for sake of time. So, let us just take a look at one simple application for this. Let us set x to be such that mu is equal to 0, and sigma is equal to the general bivariate normal, just a squared, rho a b, b square this is the general bivariate normal. Let us just set the matrix B to be the diagonal transform given by 1 by a, 0, 0, 1 by b. So, this is a diagonal transform, it is a 2 * 2 matrix, it is really corresponding to scaling the two entries of x1 and x2 by a and b.

What is the density of Y going to be? By simply applying your thing you will say Y is distributed as normal with mean given by 0,0 and the covariance metrics given by d sigma b transpose and sigma happens to be this, b happens to be this. A simple calculation will l give you 1 rho, rho 1. So, this is one reason, why we work with this particular analyze this particular density first, because even if you have a general bivariate normal just by rescaling the entries, by 1 by a and 1 by b you get back to the standard in standard bivariate normal which we analyzed.

And finally, we know that if two random variables x1 and x2 are independent then their covariance is 0. But if you also happens that x1, x2 are components of a multivariate normal, if x1 and let us say Xi and Xj are components of a multivariate normal given by x if that is the case, Xi and Xj are independent implies the covariance is 0, expected value of covariance of x1 , x2, Xi , Xj covariance of Xi , Xj which is exactly equal to sigma ij equal to 0.

We know this is true for any pair of random variables. Any pair of random variables if they are independent then their covariance is 0, we know that already, but if they also happen to be coordinates of a multivariate normal then the other direction also hooks. This is true, only this direction is true only for multivariate normal.

This direction is true for any pair of random variables. Any pair of random variables their independent implies of covariance is 0, but if they also happen to be coordinates of multivariate normal random vector then this direction also hooks. So this is pretty important to know because normal has its very special profit. So, with that we will wrap up the multivariate normal.